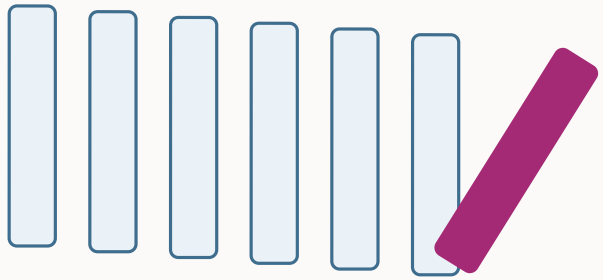
COMP233 - Discrete Mathematics

Lecture 5.2-5.3

Mathematical Induction I & II

A proof technique for infinite families of statements



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Original course materials: Prof. Mustafa Jarrar

Updated course resources: COMP233 GitHub

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Learning roadmap

What students should be able to do after these two sections

1. **Understand** why induction proves infinitely many cases
2. **Construct** basis step, inductive hypothesis, and inductive step
3. **Apply** induction to sums, divisibility, inequalities, and sequences
4. **Distinguish** mathematical induction from inductive reasoning
5. **Debug** invalid or circular induction proofs

Scope: original lecture slides + Epp 4e Sections 5.2-5.3. Strong induction (5.4) is intentionally excluded.

Why do we need mathematical induction?

The problem: proving an infinite family of statements

Finite checking is not enough

Checking P(1), P(2), P(3), ... may reveal a pattern, but no finite list can establish every remaining case.

The starting point

We prove that the first required statement P(a) is true. This is the basis step.

The propagation rule

We prove that whenever P(k) is true for an arbitrary k >= a, P(k+1) must also be true.

Think of a chain: start + reliable propagation = every case from a onward.

Inductive reasoning vs. mathematical induction

These are related ideas, but only one is a proof method

Inductive reasoning (pattern recognition)

Observe specific cases
P(1), P(2), P(3), ...

Infer a likely general pattern.

Status: useful for forming a conjecture, but not a proof.

Mathematical induction (deductive proof)

Prove a basis case P(a).

Then prove the implication P(k) -> P(k+1) for arbitrary k >= a.

Status: a valid proof of every P(n) for n >= a.

Mathematical induction itself is a deductive proof method.

Principle of Mathematical Induction

The formal engine behind the method

Claim: for every integer n >= a, P(n) is true.

1. **Basis step** — Show that P(a) is true.
2. **Inductive step** — For arbitrary k >= a, prove: if P(k) is true, then P(k+1) is true.
3. **Conclusion** — Therefore P(n) is true for every integer n >= a.

The inductive hypothesis is the assumption P(k) inside the implication - it is not a separate theorem to prove.

The induction proof template

A reusable writing structure

1. Define the property

Let P(n) be the statement you want to prove for all n >= a.

2. Basis step

Substitute n = a and verify P(a) directly.

3. Inductive step

Let k >= a be arbitrary. Assume P(k) is true (IH). Then prove P(k+1).

4. Conclusion

State explicitly that P(n) is true for every integer n >= a by mathematical induction.

Do not start by assuming P(k+1). That would be circular reasoning.

P(k) and P(k+1): write them before you calculate

Most induction errors begin here

Example property:

$$P(n): 1 + 3 + 5 + \ldots + (2n - 1) = n^2$$

$$P(k): 1 + 3 + \ldots + (2k - 1) = k^2$$

$$P(k+1): 1 + 3 + \ldots + (2k - 1) + (2k + 1) = (k+1)^2$$

→

The new term is 2(k+1)-1 = 2k+1.

Diagnostic question

If a student writes P(k+1) but leaves the last term as (2k-1), what went wrong?
Answer: they did not substitute k+1 throughout the statement.

Motivating example: 3-cent and 5-cent coins

Can every amount n >= 8 cents be formed?

P(n): n cents can be obtained using 3-cent and 5-cent coins.

Start by checking the first few amounts:

8 cents	9 cents	10 cents	11 cents	12 cents
3+5	3+3+3	5+5	3+3+5	3+3+3+3

Pattern recognition suggests the claim. The proof still requires a basis step and a valid propagation rule.

Question before the proof: how could a representation of k cents be modified to make k+1 cents?

Coin theorem: the inductive step needs cases

A good example where the step is not just algebra

Basis

P(8) is true:
8 = 3 + 5.

Inductive hypothesis

Assume k >= 8 and k cents can be formed from 3-cent and 5-cent coins.

Case 1: a 5-cent coin is used

Replace one 5-cent coin with two 3-cent coins. The total increases by 1.

Case 2: no 5-cent coin is used

Then only 3-cent coins are used. Since k >= 8, at least three 3-cent coins occur. Replace three 3s by two 5s: +1 cent.

Therefore P(k+1) is true in either case.

Sum of the first n integers

From a pattern to a proof

$$1 + 2 + 3 + \ldots + 100 = ?$$

Pair the ends:

$1+100 = 101 \quad | \quad 2+99 = 101 \quad | \quad \ldots$

The conjectured formula

For every integer n >= 1:

$1 + 2 + \ldots + n = n(n+1)/2$

Induction turns the pattern into a theorem for every n >= 1.

Worked proof: 1 + 2 + … + n = n(n+1)/2

Watch where the inductive hypothesis is used

Basis step

P(1):
1 = 1(1+1)/2 = 1.

So P(1) is true.

Inductive hypothesis

Assume, for arbitrary k >= 1:

1 + 2 + … + k = k(k+1)/2.

This is P(k).

Show P(k+1)

Start with the LHS:

1+…+k+(k+1)
= k(k+1)/2 + (k+1) [IH]
= (k+1)(k/2 + 1)
= (k+1)(k+2)/2.

That is exactly P(k+1).

Concept check: expose the previous case

The central skill in many induction proofs

Goal: prove $1 + 3 + 5 + \ldots + (2n-1) = n^2$ for $n \geq 1$.

In the inductive step, the left-hand side of P(k+1) becomes:

$$[1 + 3 + \ldots + (2k-1)] + (2k+1)$$

The bracketed part is exactly the left side of P(k), so the IH can replace it by k^2.

$$k^2 + (2k+1) = (k+1)^2$$

Strategy: rewrite P(k+1) until P(k) appears inside it.

Using the sum formula

Closed forms without re-summing term by term

Example A

Evaluate 2 + 4 + … + 500.

= 2(1+2+…+250)
= 2 * 250*251/2
= 31,375

Example B

Evaluate 5 + 6 + … + 50.

= (1+…+50) - (1+…+4)
= 50*51/2 - 4*5/2
= 1,265

Example C

For h >= 2, write 1+2+…+(h-1).

Use n = h-1:

= (h-1)h/2

Geometric sums

A second family of formulas proved by induction

For r != 1 and n >= 0: 1 + r + r^2 + ... + r^n = (r^(n+1) - 1)/(r - 1)

Example 1

1 + 3 + 3^2 + ... + 3^(m-2)

Set r = 3, n = m-2:

= (3^(m-1)-1)/2

Example 2

3^2 + 3^3 + ... + 3^m

Factor 3^2:
= 9(1+3+...+3^(m-2))
= 9(3^(m-1)-1)/2

Proof sketch for the geometric-sum formula

Induction follows the same architecture

Basis: n = 0

LHS = 1.

RHS = (r^1 - 1)/(r-1) = 1.

So P(0) is true.

Inductive hypothesis

Assume:

1+r+...+r^k = (r^(k+1)-1)/(r-1).

Show P(k+1)

Add the new term r^(k+1):

(r^(k+1)-1)/(r-1) + r^(k+1) = [r^(k+2)-1]/(r-1).

Same logic; different algebra.

Mathematics in programming

The loop and the closed form should agree

Iterative computation

```
sum = 0
for i = 1 to n:
    sum = sum + i
print(sum)
```

Closed form

```
sum = n(n+1)/2
print(sum)
```

Induction proves that both computations return the same value for every integer n >= 1.

This is an early bridge to loop invariants and algorithm correctness.

Divisibility: unpack the definition first

A useful bridge back to Chapter 4

$$a \mid b \text{ means } b = aq \text{ for some integer } q.$$

When the inductive hypothesis says “3 divides 2^(2k)-1”, immediately rewrite it as:

$$2^{(2k)} - 1 = 3r \text{ for some integer } r.$$

Why this matters

The algebraic form 3r gives you something concrete to substitute into P(k+1). Definitions are proof tools, not vocabulary to memorize.

Worked proof: 3 divides 2^(2n)-1

For every integer n >= 0

Basis

n = 0:
2^0 - 1 = 0,
and 3 | 0.

So P(0) is true.

IH

Assume:
2^(2k)-1 is divisible by 3.

So 2^(2k)-1 = 3r for some r in Z.

Show P(k+1)

2^(2(k+1)) - 1
= 4*2^(2k) - 1
= 3*2^(2k) + (2^(2k)-1)
= 3*2^(2k) + 3r
= 3(2^(2k)+r).

The factor in parentheses is an integer, so divisibility follows.

Worked proof: an inequality

Prove 2n + 1 < 2^n for every integer n >= 3

Basis

n = 3:
2(3)+1 = 7 < 8 = 2^3.

P(3) is true.

IH

Assume k >= 3 and

2k + 1 < 2^k.

We must show:
2(k+1)+1 < 2^(k+1).

Inductive step

2k+3 = (2k+1)+2
< 2^k + 2
< 2^k + 2^k
= 2^(k+1).

Why is 2 < 2^k? Because k >= 3.

Induction with inequalities: what changes?

The logic is the same, but every inequality step needs justification

Safe moves

Add the same quantity to both sides.

Multiply by a positive quantity.

Use a known bound that follows from k >= a.

Dangerous moves

Multiplying by an expression of unknown sign.

Using a bound true only for small k.

Claiming “obvious” without linking it to k >= a.

Instructor question

Where did the inductive hypothesis stop, and where did ordinary inequality reasoning begin?

A recursively defined sequence

From observed terms to a closed form

$$a1 = 2, \quad ak = 5a(k-1) \text{ for } k >= 2$$

a1	a2	a3	a4
2	10	50	250

The pattern suggests:

$$an = 2 * 5^{(n-1)}$$

Now prove it - do not stop at the pattern.

Proof of the sequence formula

Use the recurrence inside the inductive step

Basis

a1 = 2.

Formula gives:
2*5^(1-1) = 2.

So P(1) is true.

IH

Assume:

ak = 2*5^(k-1).

This is P(k).

Show P(k+1)

a(k+1) = 5ak
= 5[2*5^(k-1)] [IH]
= 2*5^k
= 2*5^((k+1)-1).

Therefore P(k+1) is true.

Always verify the basis case directly from the sequence definition.

Deduction, inductive reasoning, and mathematical induction

Three distinct reasoning patterns

Deductive reasoning

Starts from accepted facts, definitions, or theorems and derives a logically necessary conclusion.

Example: All humans are persons; Sami is human; therefore Sami is a person.

Inductive reasoning

Uses observed cases or experience to form a plausible generalization.

Example: recent lectures had quizzes; conjecture that the next lecture will have one.

Mathematical induction

A deductive proof principle for statements indexed by integers.

It proves a basis case and the propagation implication P(k)->P(k+1).

Proof debugging: circular reasoning

Concept Check

A proof can look formal and still be invalid

Invalid step

“Assume P(k+1) is true.
Then after some algebra, P(k+1) is true.”

Problem: the desired conclusion was assumed at the start.

Valid structure

Assume only P(k).

Use definitions, algebra, and previously proved facts to derive P(k+1).

The proof must visibly depend on the IH.

Diagnostic question: “Where exactly did you use P(k), and did you assume any part of P(k+1)?”

Common misconceptions and quick fixes

What to watch for in your own proof

Typical error	Correction
Trying to prove P(k)	P(k) is the inductive hypothesis inside an implication.
Checking many cases	Examples suggest a pattern; they do not prove all n.
Wrong P(k+1)	Substitute k+1 everywhere before simplifying.
Skipping the basis	The chain has no starting point.
Never using the IH	Then the inductive step may not actually connect cases.
Using a wrong starting index	The basis and the range k >= a must match the theorem.

Concept check 1: identify the structure

Practice

Do not calculate yet

Claim: for all $n >= 1$, $2 + 4 + \ldots + 2n = n(n+1)$.

A. Write P(k)

What statement are you allowed to assume in the inductive step?

B. Write P(k+1)

What is the new final term when n becomes k+1?

C. Strategy

How can the left-hand side of P(k+1) be split so that P(k) appears?

Think for 90 seconds before revealing a solution.

Concept check 2: valid or invalid?

Practice

Find the earliest failing point

Claim

For all n >= 1, n^2 >= n.

“Proof”: Assume k^2 >= k. Then (k+1)^2 >= k+1. Therefore the statement is true for all n >= 1.

Questions

1. Is the inductive step valid?
2. What is missing?
3. Can the missing part be repaired?

Hint: induction requires both a starting point and propagation.

Learning Appendix

Core concepts • connections • worked examples • knowledge check • answers • enrichment

Use this section for revision after the lecture.

Appendix A - Core concepts

A one-slide revision sheet

Proof skeleton

1. Define P(n) and start a.
2. Prove P(a).
3. Let k >= a be arbitrary.
4. Assume P(k).
5. Prove P(k+1).
6. Conclude by induction.

Key conditions

The index is integer-valued.

The basis matches the claimed range.

The inductive step must work for every allowed k.

The IH may be used - not proved.

Common proof types

Equalities and sums

Geometric sequences

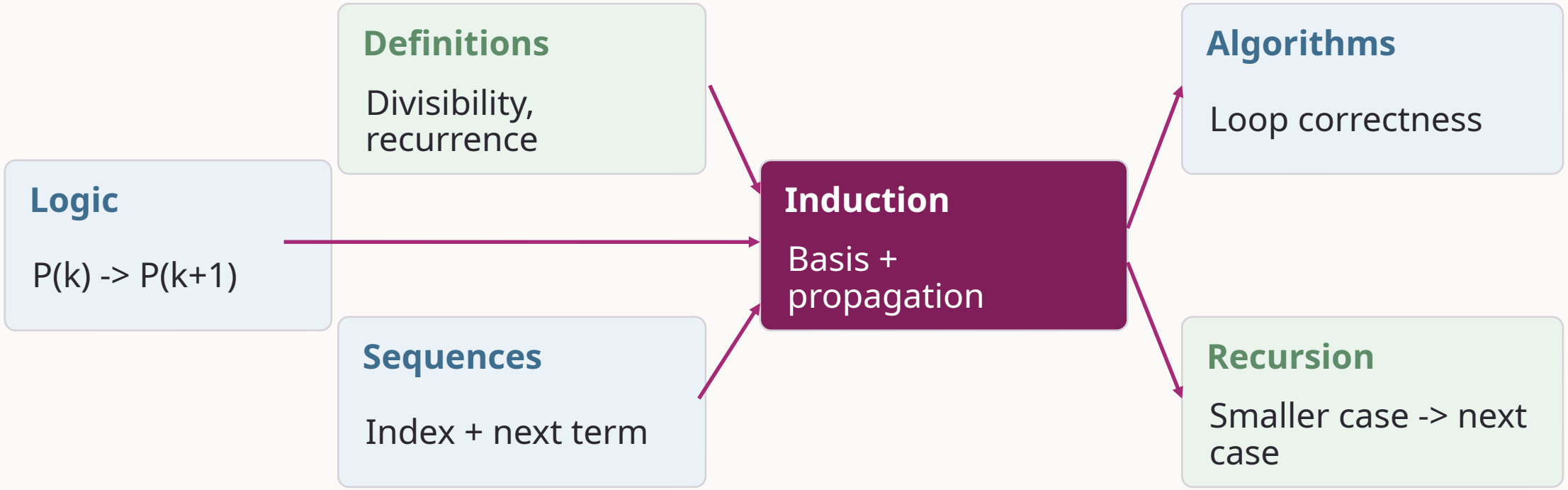
Divisibility

Inequalities

Properties of recursively defined sequences

Appendix B - Conceptual connections

How induction links earlier and later ideas



Induction is not isolated: it reuses logic and definitions, then prepares students for algorithmic and recursive reasoning.

Appendix C - Worked examples I

A valid example and a boundary check

Valid: 1 + 3 + ... + (2n-1) = n^2

Basis n=1: 1=1.

IH: 1+3+...+(2k-1)=k^2.

Step:
LHS(k+1)=k^2+(2k+1)=(k+1)^2.

Therefore true for all n>=1.

Boundary: why does 2n+1 < 2^n start at n=3?

n=1: 3<2 is false.
n=2: 5<4 is false.
n=3: 7<8 is true.

So the theorem cannot start earlier than 3.

The basis index is mathematically meaningful, not cosmetic.

Appendix C - Worked examples II

Invalid proof patterns to recognize

Invalid pattern A: missing basis

“Assume P(k). Show P(k+1). Therefore P(n) for all n.”

Why invalid?
There may be no starting case connected to the chain.

Invalid pattern B: checking examples

“P(1), P(2), …, P(1000) are true, so P(n) is true for all n.”

Why invalid?
A counterexample could occur at n=1001 or later.

Appendix D - Knowledge Check 1

Conceptual questions

Question 1

1. MCQ: In an inductive step, what may you assume?
A) P(k+1) B) P(k) C) the theorem is true for all n D) the basis is unnecessary

Question 2

2. True/False with justification: Verifying the first 10,000 cases proves a universal statement over all integers.

Question 3

3. Write P(k) and P(k+1) for: $2+4+\ldots+2n = n(n+1)$.

Appendix D - Knowledge Check 2

Application and error detection

Question 4

4. Prove by induction: 2+4+...+2n = n(n+1) for n>=1.

Question 5

5. Error detection: a proof assumes P(k+1) in the inductive step and later derives P(k+1). Identify the exact logical error.

Question 6

6. Counterexample: Find the smallest n showing that 2n+1 < 2^n cannot hold for all n>=1.

Appendix D - Knowledge Check 3

Higher-order reasoning

Question 7

7. Explain why the basis case and inductive step together are sufficient to prove infinitely many cases.

Question 8

8. Prove by induction that 3 divides 2^(2n)-1 for n>=0, explicitly unpacking the divisibility definition.

Question 9

9. A sequence is defined by a1=3 and a(k)=7a(k-1). Conjecture a closed form and prove it by induction.

Appendix E - Answer Key 1

Questions 1-3

1. MCQ

Answer: B, P(k).

A is circular. C assumes the conclusion. D removes the starting point.

2. True/False

False. Finite checking can support a conjecture but cannot establish all remaining integer cases.

3. P(k), P(k+1)

P(k): $2+4+\ldots+2k=k(k+1)$.

P(k+1): $2+\ldots+2k+2(k+1)=(k+1)(k+2)$.

Appendix E - Answer Key 2

Questions 4-6

4. Even-number sum

Basis n=1: 2=1*2.

IH: 2+...+2k=k(k+1).

Step: add 2(k+1):
k(k+1)+2(k+1)=(k+1)(k+2).

5. Error

The proof assumes the target P(k+1), so it is circular.

Only P(k) may be assumed.

6. Counterexample

n=1 already fails:
2(1)+1 = 3, but 2^1 = 2.

Also n=2 fails. The theorem begins at n=3.

Appendix E - Answer Key 3

Questions 7-9

7. Why it works

P(a) starts the chain.

The implication P(k)->P(k+1) guarantees each true case forces the next.

Therefore every integer case from a onward is reached.

8. Divisibility

Use the proof from Slide 18.
The key conversion is:
2^(2k)-1 = 3r, r in Z,
then factor 3 from the k+1 expression.

9. Sequence

Conjecture: an = 3*7^(n-1).

Basis: a1=3.
IH: ak=3*7^(k-1).
Then a(k+1)=7ak=3*7^k.

Appendix F - Optional enrichment

From induction to algorithm correctness

Correct after k iterations -> correct after k+1 iterations

Loop invariants

A property that remains true before and after each loop iteration often has an induction-like proof structure.

Recursive algorithms

Correctness of a recursive step often depends on correctness for a smaller input - another induction pattern.

Cybersecurity / AI

Induction can verify iterative state updates, recursive search procedures, and properties of generated structures.

References and course resources

Primary mathematical reference

Susanna S. Epp, Discrete Mathematics with Applications, 4th ed., Chapter 5, Sections 5.2-5.3.

Original course materials

Prof. Mustafa Jarrar - Discrete Mathematics course materials. Dr. Radi Jarrar is acknowledged for subsequent course-material contributions.

Original course materials	https://www.jarrar.info/courses/DMath/
Updated COMP233 resources	https://github.com/RuwaYafa/BZU_Courses/tree/main/COMP233

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